

Eight structural principles → architectural commitments

Principle

Architectural commitment

■■ Established physical, informational, and statistical laws (external references) ■■

1. Second Law of Thermodynamics

Entropy treated as architectural diagnostic, not fault

2. Ashby's Law of Requisite Variety

Fleet of specialist agents covering operational subdomains

3. Shannon Information Theory

Variance reduction at the signal-carrier level

4. Principle of Least Action

Explicit-target single-dispatch no implicit resolvers

5. Lyapunov-style bounded response

Halt-and-resume pathway through audited bus

6. Power-Law distribution (empirical)

Engineered for the 99th-percentile event, not the median

■■ Original frameworks proposed by the author (design-stage; pending empirical validation) ■■

7. Individual-Baseline Variance Modeling*

Operator-baseline calibration; no population-norm comparison

8. Precision Perturbation Without Variance Compression*

Probability-mass intervention preserving operator authority

* Original frameworks introduced for the operator-cognitive performance layer; full operational specification is forthcoming work.